

STAGE 1

Input and node encoders

Terrain nodes
 $x_t \in \mathbb{R}^{18}$
12.96 M nodes

Terrain encoder
 $18 \rightarrow 256 \rightarrow 256$
BN + LeakyReLU(0.2)

Antenna nodes
 $x_a \in \mathbb{R}^6$
175-4252 nodes

Antenna encoder
 $6 \rightarrow 128 \rightarrow 128 \rightarrow 256$
BN + LeakyReLU(0.2)

STAGE 2

Heterogeneous message passing, $\times 4$ layers

one HeteroConv layer (sum over relations)

GATv2 antenna $\rightarrow$ terrain
4 heads $\times$ 64, edge_dim = 2

GATv2 terrain $\rightarrow$ terrain
4 heads $\times$ 64, edge_dim = 2

SAGEConv terrain $\rightarrow$ antenna
mean aggregation

LeakyReLU $\cdot$ Dropout(0.1) $\cdot$ residual $\cdot$ LayerNorm

terrain and antenna embeddings $h \in \mathbb{R}^{256}$

STAGE 3

Physics-constrained decoder

MLP
 $256 \rightarrow 256$
 $\rightarrow 128 \rightarrow 5$
GELU +
Dropout

L_{total} : softplus, clamp [0, 200] dB

L_{veg} : softplus, clamp [0, 50] dB

L_{ter} : softplus, clamp [0, 30] dB

RSSI: $-150 + 150 \sigma(\cdot)$ dBm

p_{cov} : $\sigma(\cdot) \in [0, 1]$

*free-space floor, distance monotonicity, variance and NDVI sign terms
are penalised in the loss and verified at the output*